(ii) Division by 10, 100 and 1000

Let us learn how to divide decimal numbers by 10, 100 and 1000.

For example, consider $51.7 \div 10$

Now $51.7 = \frac{517}{10}$

Therefore, $\frac{51.7}{10} = \frac{517}{10} \times \frac{1}{10} = \frac{517}{100} = 5.17$

Similarly, $51.7 \div 100 = \frac{517}{10} \times \frac{1}{100} = \frac{517}{1000} = 0.517$

Also $51.7 \div 1000 = \frac{517}{10} \times \frac{1}{1000} = \frac{517}{10000} = 0.0517$

When decimal numbers are divided in powers of 10, it can be noted that the decimal number so obtained contains the same number of decimal digits as that of zeros in powers of 10.

Let us see whether there is a pattern for dividing numbers by 10, 100 and 1000.

Observe the following and complete it.

Try these

$36.7 \div 10 = 3.67$	$436.7 \div 10 = 43.67$	$2.3 \div 10 = 0.23$	$27.17 \div 10 =$ ____
$36.7 \div 100 = 0.367$	$436.7 \div 100 =$ _____	$2.3 \div 100 =$ _____	$27.17 \div 10 =$ _____
$36.7 \div 1000 = 0.0367$	$436.7 \div 1000 =$ _____	$2.3 \div 1000 =$ _____	$27.17 \div 1000 =$ _____

Take. $36.7 \div 10 = 3.67$. In 36.7 and 3.67, the digits are the same namely, 3, 6 and 7, but the decimal point has been shifted to the left by one place in the obtained decimal number after division. Note that 10 is one zero followed by 1.

Similarly, in $36.7 \div 100 = 0.367$, the decimal point has been shifted to the left by two places in the obtained decimal number after division. Note that 100 is two zeros followed by 1.

Similarly, in $36.7 \div 1000 = 0.0367$, the decimal point has been shifted to the left by three places in the obtained decimal number after division. Note that 1000 is three zeros followed by 1.

Try these

Divide the following

(i) $17.237 \div 10$

(ii) $17.237 \div 100$

(iii) $17.237 \div 1000$

Hence we can conclude that while dividing a decimal number by 10, 100 and 1000, the digits of the number (Dividend) and the obtained decimal number after division are the same but the decimal point in the obtained decimal number after division is shifted to the left by as many places as there are zeros followed by 1.

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